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1 Introduction**2 Methodology***2.1 Experimental Setup*

2.1.1 Multi-camera configuration A new experimental setup was designed based on the configuration proposed by [?], incorporating modifications aimed at improving measurement repeatability, image quality, and spatial resolution. A multi-camera system was implemented to simultaneously capture both global flow patterns and localized high-speed phenomena. Figure 1 illustrates the camera distribution around the L-shape and beam containers.

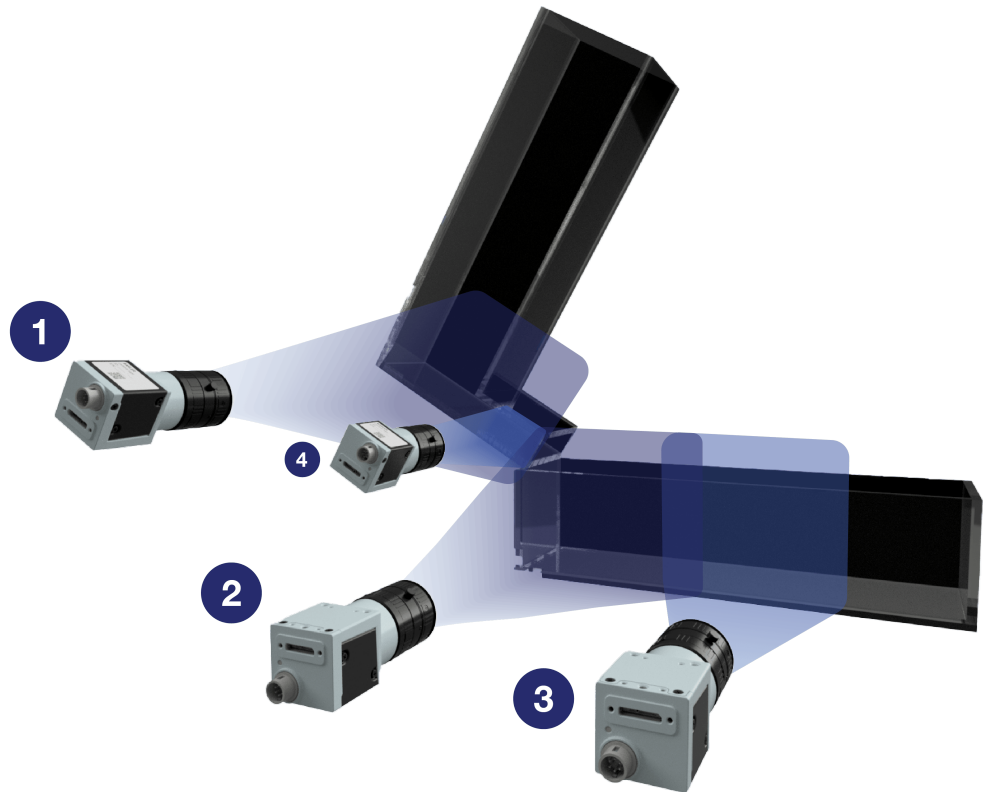


Figure 1: Multi-camera setup for simultaneous PIV and PTV measurements. Cameras 1–3 provide wide-field coverage of the L-shape container (Cam 1) and beam container from orthogonal views (Cams 2–3). Camera 4 captures high-speed close-up footage of the L-shape outlet region to resolve initial flow development.

Camera 1 captures the region of interest (ROI) of the L-shape container to track overall flow progression and fiber distribution, while Cameras 2 and 3 record the beam container from orthogonal angles to enable stereoscopic reconstruction of three-dimensional fiber orientations. Camera 4 was positioned to capture a close-up view of the L-shape outlet at a higher frame rate, resolving the rapid fiber reorientation occurring during the transition from vertical to horizontal flow. Table ?? summarizes the specifications of each camera.

All cameras were equipped with 24 mm $f/1.8$ lenses. The wide aperture ($f/1.8$) enables short exposure times, reducing motion blur, at the cost of a reduced depth of field. The shallow depth of field is acceptable because the planar laser sheet for PIV and the narrow measurement volume for PTV ensure that only features within the focal plane are illuminated and contribute to the measurements.

Table 1: Camera specifications and operating parameters.

Camera	Frame Rate (fps)	Resolution (pixels)	Scale Factor (mm/px)
1	220	1024 × 1024	0.1250
2	220	1024 × 1024	0.1282
3	220	1024 × 1024	0.1282
4	660	384 × 384	0.0935

2.1.2 *Illumination for PIV and PTV* **HABLAR DE LOS TIPOS DE ILUMINACION USADAS PARA PIV Y PTV, Y JUSTIFICAR LA ELECCION DE CADA UNA.**

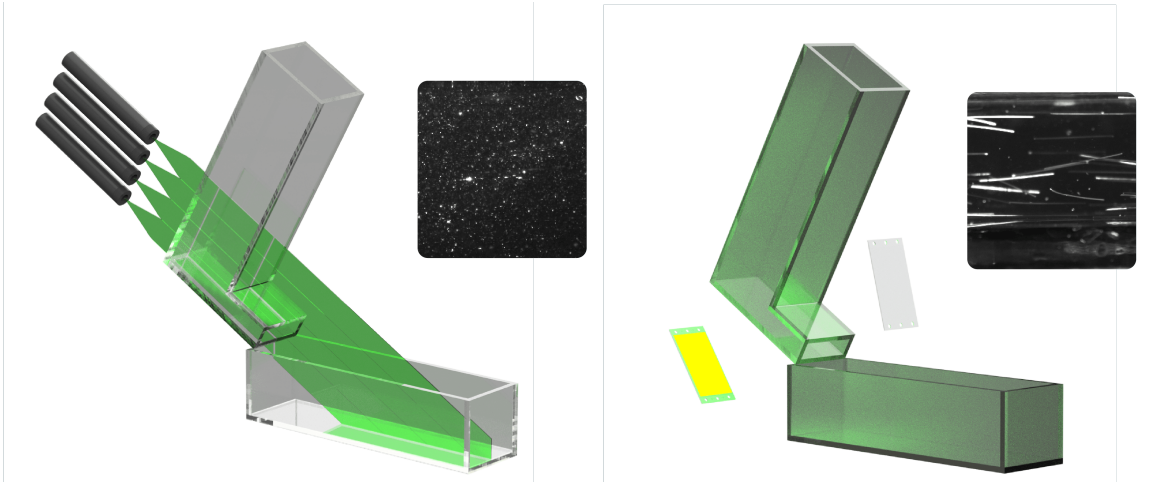


Figure 2: **COMPLETAR**

2.1.3 *Experimental procedure* The semi-dilute concentration limit was used as the primary reference for defining the fiber concentrations. For each Carbopol batch prepared (2 L total volume), a volume loss of 198 mm³ per fiber was estimated, yielding a final suspension volume of 1.802 L. At this volume, the semi-dilute limit of 0.024% volume fraction [?, ?] corresponds to approximately 1023 fibers in the Carbopol batch. Since the L-shaped device was filled only up to a height of 22.4 mm, this translates to approximately 746 fibers effectively deposited within the measurement domain. Based on this calculation, one fiber concentration below the semi-dilute limit and two above it were selected, as summarized in Table ??.

Fibers were counted by weighing them on a precision balance whose resolution was at least one order of magnitude smaller than the weight of a single fiber, ensuring reliable individual fiber counts. The resulting concentrations, expressed in terms of number of fibers, weight fraction, and volume fraction, are summarized in Table ??.

Each concentration was tested using two Carbopol formulations (0.2% and 0.5%), to assess the sensitivity of the results to batch-to-batch rheological variability. For each formulation-concentration combination, two independent batches were prepared, and each batch was used for two experimental

Table 2: Fiber concentrations and corresponding measurement configurations used for CFD-DEM validation. The number of fibers in L refers to those deposited within the L-shaped device at a fill height of 22.4 mm.

N° of Fibers	Weight Fraction	Volume Fraction	N° of Fibers in L	Method
0	0.000%	0.000%	0	PIV only
750	0.135%	0.017%	547	PIV & PTV
1023	0.185%	0.024%	746	Semi-dilute
1500	0.271%	0.035%	1094	PIV & PTV
3000	0.541%	0.069%	2188	PIV & PTV

runs, yielding a total of 16 runs per Carbopol type. In experiments with fibers, PTV was performed first to record fiber dynamics; tracer particles were subsequently added to the same batch to perform PIV and characterize the velocity field under identical flow conditions. In fiber-free experiments, two PIV runs were performed per batch. Across all tests per formulation, this protocol resulted in 10 PIV datasets and 6 PTV datasets per formulation.

Figure ?? shows representative images of the L-shaped device prior to each experimental run, illustrating the progressive increase in fiber density across the tested concentrations. These images confirm the visual distinction between loading conditions and provide qualitative context for the quantitative tracking results discussed in Section ??.



Figure 3: Representative images of the L-shaped device prior to experimental runs at each fiber concentration: 750, 1500, and 3000 fibers, shown for both Carbopol-02 and Carbopol-05 formulations. Images illustrate the progressive increase in fiber density and spatial overlap as fiber loading increases.

2.2 Steel Fibers

The steel fibers used in all experiments were Dramix[®] OL 13/.20 (Bekaert), a commercially available straight steel fiber designed for fiber-reinforced concrete applications [?]. Their physical and mechanical properties are summarized in Table ??.

Table 3: Physical and mechanical properties of the Dramix[®] OL 13/.20 steel fibers used in the experiments.

Property	Value	Unit
Total length (L)	13.000	mm
Diameter (d)	0.200	mm
Aspect ratio (L/d)	65.000	–
Elastic modulus	200.000	GPa
Tensile strength	2.750	GPa
Numerical density	282.556	fibers/kg

2.3 Carbopol Preparation

Carbopol Ultrez 10 (Lubrizol) was selected as a transparent viscoplastic analog for fresh UHPC, following the approach established in prior works [?, ?]. Its optical transparency allows direct visualization of fibers and tracer particles, while its stable, time-independent rheology eliminates the hydration effects that complicate testing with real cementitious materials [?]. Two concentrations were prepared: 0.2% and 0.5% (Carbopol/water by mass).

For each batch, deionized water was first measured and its initial pH verified to be approximately 7.0. The corresponding amount of Carbopol powder was gradually added over one minute while mixing with a mechanical stirrer at 150 rpm; stirring was maintained for one hour to ensure complete and homogeneous polymer dispersion. Following dispersion, the pH was measured at five points throughout the mixture. The first dose of 1 N NaOH solution was then added and the mixture was stirred for 30 min. After this period the pH was measured again, the second NaOH dose was added, and stirring continued for an additional 30 min. A final five-point pH measurement confirmed that the average value fell within the accepted range. The specific quantities for each concentration, together with the target pH range, are summarized in Table ??.

Table 4: Preparation parameters for Carbopol Ultrez 10 solutions at 0.2% and 0.5% concentrations. NaOH additions are sequential: the primary dose initiates gelation and the secondary dose achieves the target pH range.

Carbopol	Water (mL)	Carbopol (g)	NaOH ₁ (mL)	NaOH ₂ (mL)	Target pH
0.2%	1996	4.0	26	3	6.8–7.2
0.5%	1990	10.0	70	20	6.8–7.2

The resulting gel was stored in a sealed container at 20 °C for 24 h to ensure complete hydration and structural stabilization. Once stabilized, each batch was centrifuged in a Hettich Universal 320R centrifuge at 2000 rpm and 20 °C for 10 min to remove entrapped air bubbles, yielding a clear, optically homogeneous gel suitable for flow visualization [?]. Each prepared batch was used for a maximum of two experimental runs, and the same protocol was applied to both PIV and PTV measurements.